

Chapter 4: Basics of Probabilistic Modelling

Discriminative vs. Generative Modeling – logistic regression: Formulate the logistic model,

$$X \sim f(x) \text{ and } Y|X = x \sim \text{Ber}(\mu_\theta(x))$$

where $\mu_\theta(x) = \text{sig}(\beta_0 + \beta_1 x)$ and $\theta = (\beta_0, \beta_1)^T$, and describe how to find the MLE.

Discriminative vs. Generative Modeling – linear discriminant analysis: Formulate the linear discriminant analysis (LDA),

$$Y \sim \text{Ber}(p) \text{ and } X | Y = y \sim \mathcal{N}(\mu_y, \sigma^2)$$

and describe how to find the MLE.

Discriminative vs. Generative Modeling: Compare the two paradigms.

Curse of Dimensionality: Understand how Naïve Bayes and multivariate LDA model within-feature dependencies, and examine their limitations in high-dimensional settings.

Dimensionality Reduction – Linear Discriminant Analysis: Understand that LDA finds the projection axis that maximizes the Mahalanobis distance between two classes.

Dimensionality Reduction – Principal Component Analysis: Understand that PCA sequentially finds the axes that maximize explainable variance under the orthonormality condition.

Dimensionality Reduction – Independent Component Analysis: Derive independent components to identify statistically independent embeddings.

Dimensionality Reduction – Autoencoders: Understand that autoencoders can be interpreted as a non-linear extension of PCA.